

UNIT 4: MOLECULAR GENETICS

4.9 – MENDELIAN INHERITANCE

PART 2 – GENETIC DISORDERS

Mendelian Inheritance Outline

- Part 1 – Mendelian Genetics
 - Gregor Mendel
 - Mendel's Laws
- Part 2 – Genetic Disorders
 - Extending the Range of Mendellian Inheritance

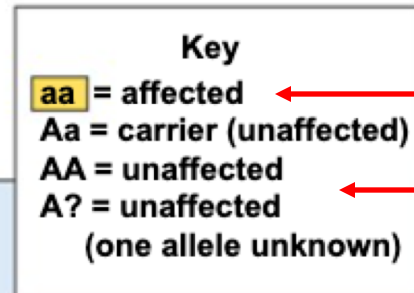
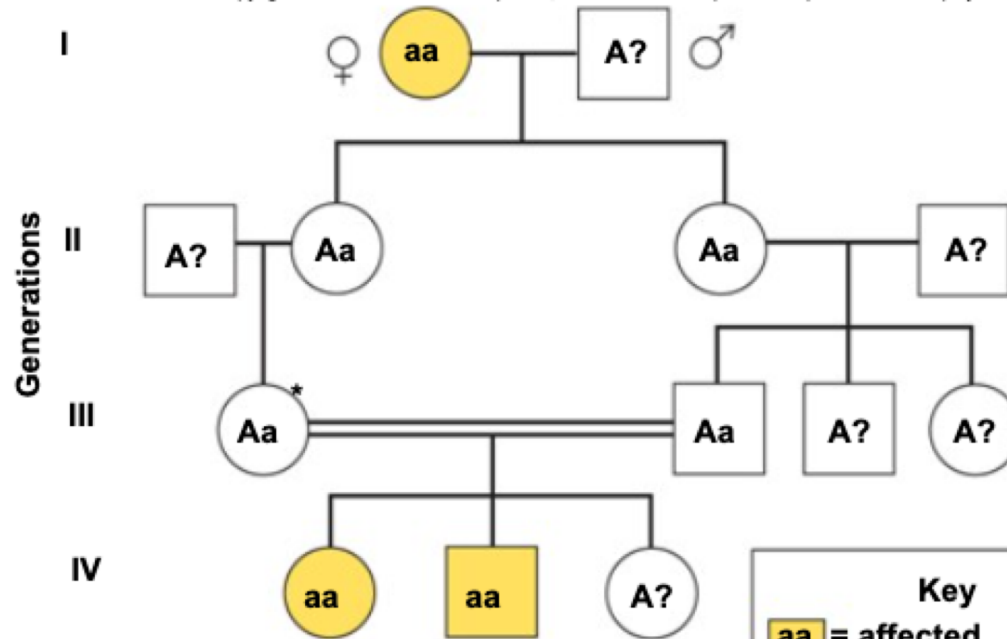
Mendel's Laws

- Genetic disorders are medical conditions caused by alleles inherited from parents
- **Autosome** - Any chromosome other than a sex chromosome (X or Y)
- Genetic disorders caused by genes on autosomes are called **autosomal disorders**
 - ▣ Some genetic disorders are **autosomal dominant**
 - An individual with AA has the disorder
 - An individual with Aa has the disorder
 - An individual with aa does NOT have the disorder
 - ▣ Other genetic disorders are **autosomal recessive**
 - An individual with AA does NOT have the disorder
 - An individual with Aa does NOT have the disorder, but is a carrier
 - An individual with aa DOES have the disorder

Pedigree

- A **pedigree** is like an allele Family Tree
 - ▣ Males represented by squares
 - ▣ Females represented by circles
- ▣ Unions between two individuals marked by a horizontal line between them
- ▣ Resulting offspring marked by a vertical line
- ▣ Multiple offspring line up on the same horizontal plane
- ▣ Method of tracing patterns of inheritance and tracing disease states and type of disorder

Autosomal Recessive Pedigree



Autosomal recessive disorders

- Most affected children have unaffected parents.
- Heterozygotes (Aa) have an unaffected phenotype.
- Two affected parents will always have affected children.
- Close relatives who reproduce are more likely to have affected children.
- Both males and females are affected with equal frequency.

Autosomal Recessive Disorders

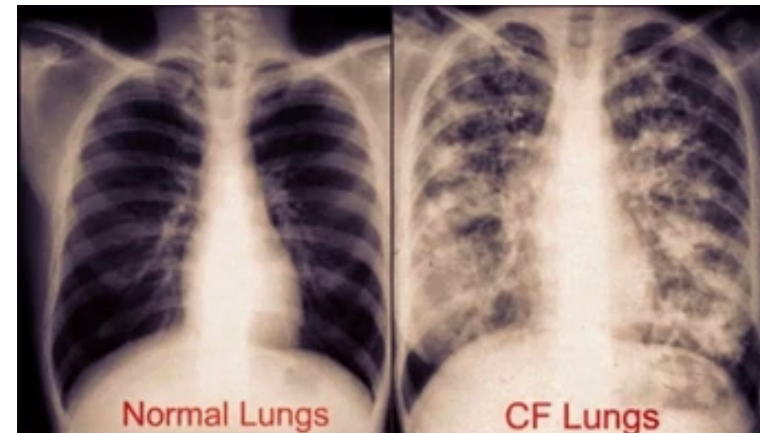
□ Methemoglobinemia

- Relatively harmless disorder
- Accumulation of methemoglobin in the blood causes skin to appear bluish-purple

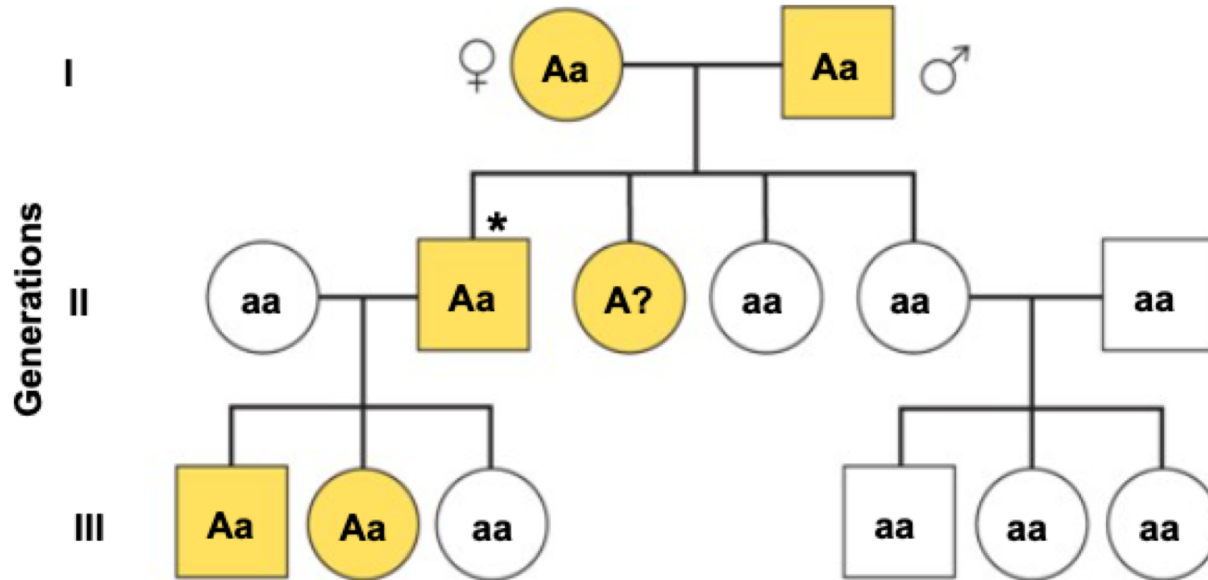


□ Cystic Fibrosis

- Mucus in bronchial tubes and pancreatic ducts is particularly thick and viscous



Autosomal Dominant Pedigree



Autosomal dominant disorders

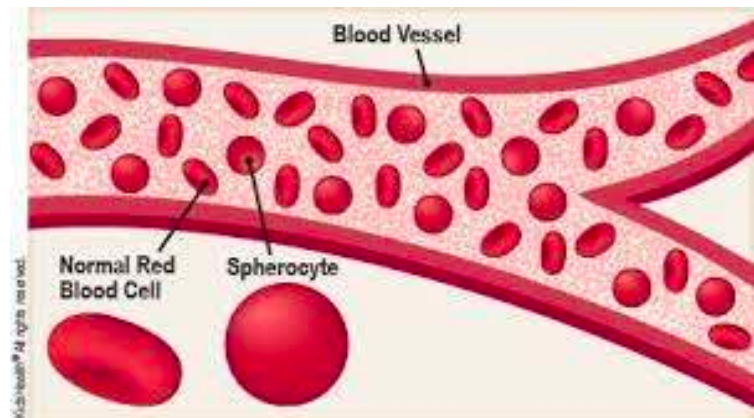
- Affected children will usually have an affected parent.
- Heterozygotes (Aa) are affected.
- Two affected parents can produce an unaffected child.
- Two unaffected parents will not have affected children.
- Both males and females are affected with equal frequency.

Key

AA = affected
Aa = affected
A? = affected
(one allele unknown)
aa = unaffected

Autosomal Dominance Disorders

- Osteogenesis Imperfecta
 - Weakened, brittle bones
 - Most cases are caused by mutation in genes required for the synthesis of type I collagen
- Hereditary Spherocytosis
 - Caused by a mutation in the ankyrin-1 gene
 - Red blood cells become spherical, are fragile, and burst easily



Extending the Range of Mendelian Genetics

- Some traits are controlled by **multiple alleles** (multiple allelic traits)
- The gene exists in several allelic forms (but each individual only has two alleles)
- ABO blood types
 - The alleles:
 - I^A = A antigen on red blood cells, anti-B antibody in plasma
 - I^B = B antigen on red blood cells, anti-A antibody in plasma
 - i = Neither A nor B antigens on red blood cells, both anti-A and anti-B antibodies in plasma
- The ABO blood type is also an example of **codominance**
 - More than one allele is fully expressed
 - Both I^A and I^B are expressed in the presence of the other

Extending the Range of Mendelian Genetics

Phenotype

A

B

AB

O

Genotype

$I^A I^A, I^A i$

$I^B I^B, I^B i$

$I^A I^B$

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Extending the Range of Mendelian Genetics

□ Incomplete Dominance:

- Heterozygote has a phenotype intermediate between that of either homozygote
 - Homozygous red has red phenotype
 - Homozygous white has white phenotype
 - Heterozygote has pink (intermediate) phenotype
- Phenotype reveals genotype without a test cross

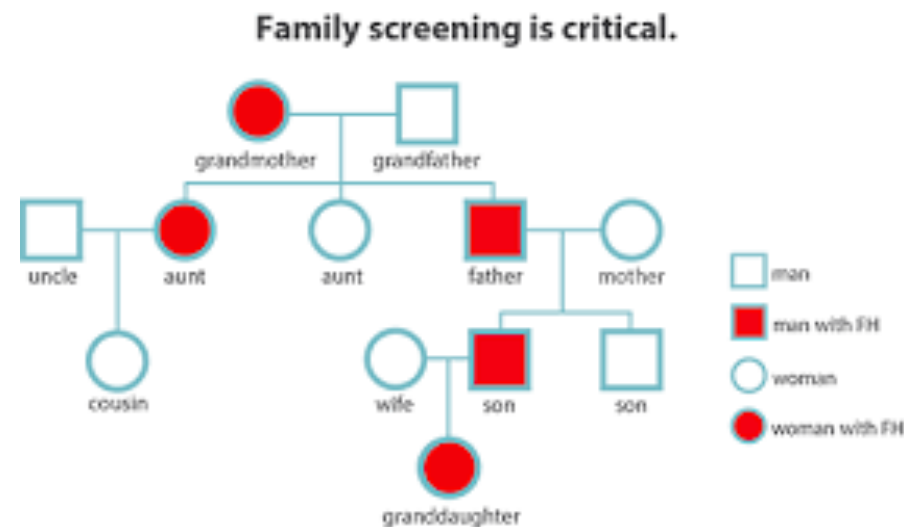


Extending the Range of Mendelian Genetics

□ Human examples of incomplete dominance:

▣ Familial Hypercholesterolemia (FH)

- Homozygotes for the mutant allele develop fatty deposits in the skin and tendons and may have heart attacks during childhood
- Heterozygotes may suffer heart attacks during early adulthood
- Homozygotes for the normal allele do not have the disorder



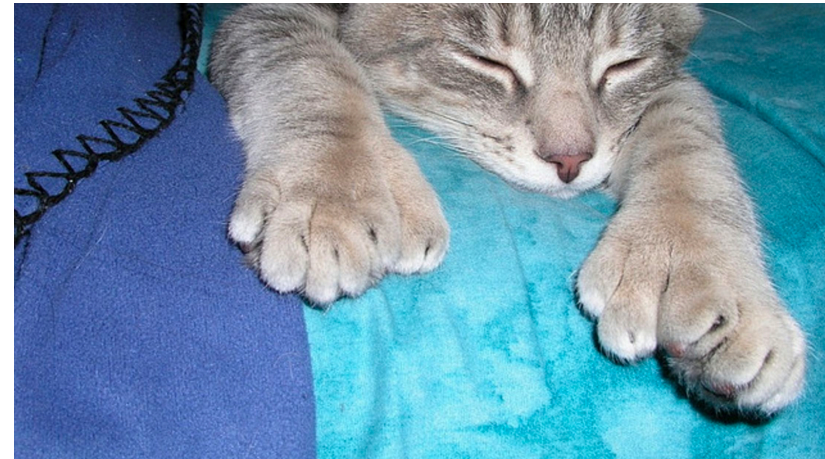
*Courtesy of the FH Foundation

Extending the Range of Mendelian Genetics

□ Human examples of incomplete dominance:

■ **Incomplete penetrance**

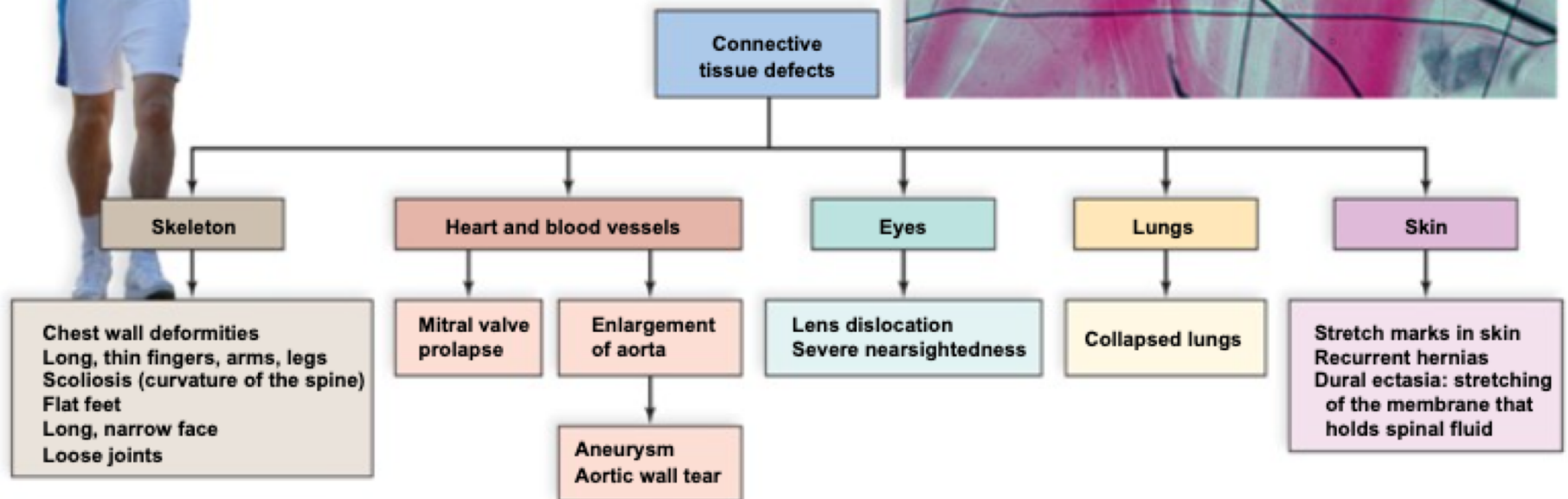
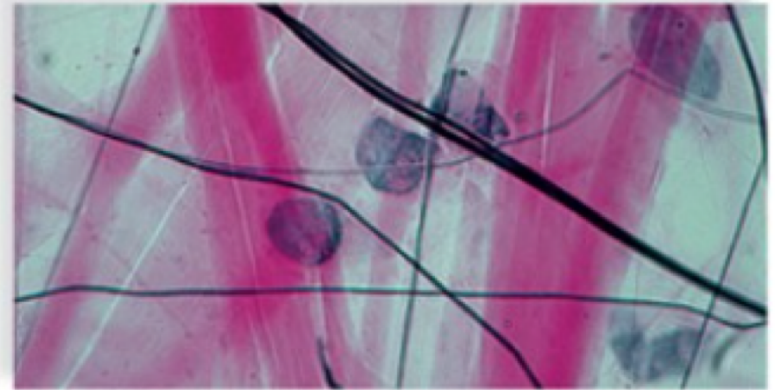
- The dominant allele may not always lead to the dominant phenotype in a heterozygote
- Many dominant alleles exhibit varying degrees of penetrance
- Example: polydactyly
 - Extra digits on hands, feet, or both
 - Not all individuals who inherit the dominant polydactyly allele will exhibit the trait



Extending the Range of Mendelian Genetics

- **Pleiotropy** occurs when a single mutant gene affects two or more distinct and seemingly unrelated traits.
- Marfan syndrome is pleiotropic and results in the following phenotypes:
 - ▣ disproportionately long arms, legs, hands, and feet
 - ▣ a weakened aorta
 - ▣ poor eyesight
 - All seemingly unrelated, but due to the production of abnormal connective tissue resulting from a mutated fibrillin gene on chromosome 15 (*FBN1*)
- Sickle cell Anemia another example

Extending the Range of Mendelian Genetics



Extending the Range of Mendelian Genetics

□ **Polygenic Inheritance:**

- Occurs when a trait is governed by two or more genes having different alleles
- Each dominant allele has a quantitative effect on the phenotype
- These effects are additive
- Results in continuous variation of phenotypes within a population
- The traits may also be affected by the environment
- Examples
 - Human skin color
 - Height
 - Eye color
- Disorders known to involve multiple genes having an effect include cleft lip, hypertension, diabetes, allergies and cancer

Extending the Range of Mendelian Genetics

Skin Color

Genotypes

AABB

AABb or *AaBB*

AaBb or *AAbb* or *aaBB*

Aabb or *aaBb*

aabb

Phenotype

Very dark

Dark

Medium Brown

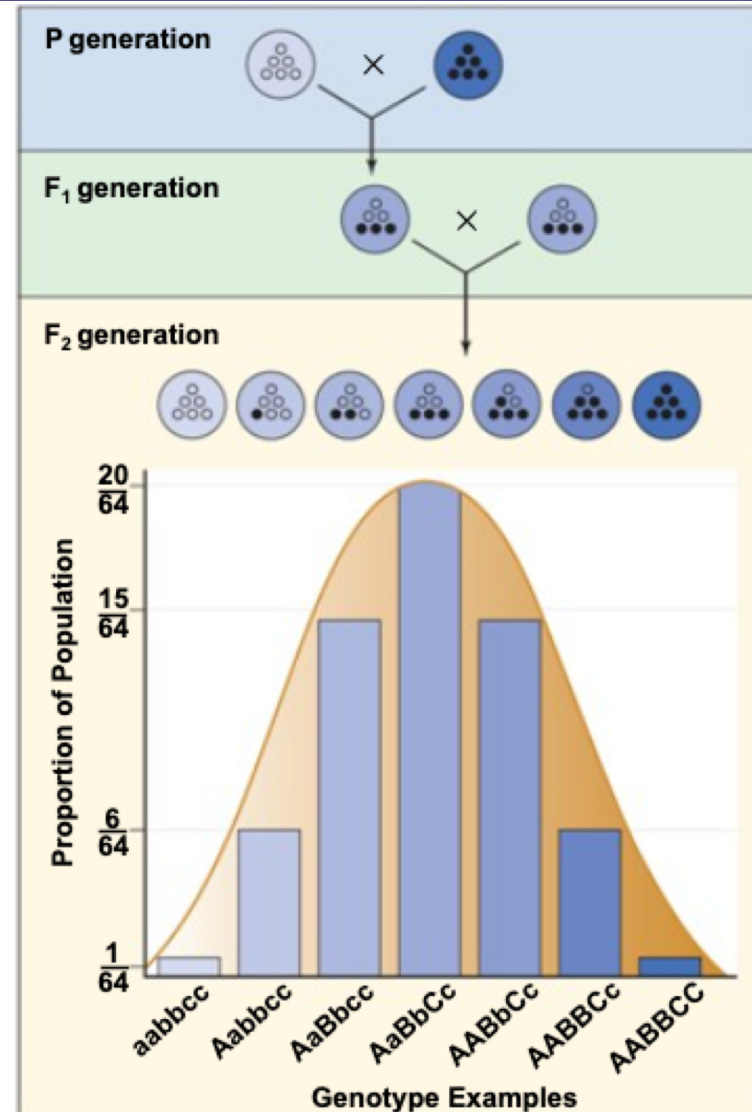
Light

Very Light

Chart Key

Black dots and intensity of blue shading stand for # of dominant alleles

Orange shading represents degree of environmental influence



Extending the Range of Mendelian Genetics

□ **X-Linked Inheritance**

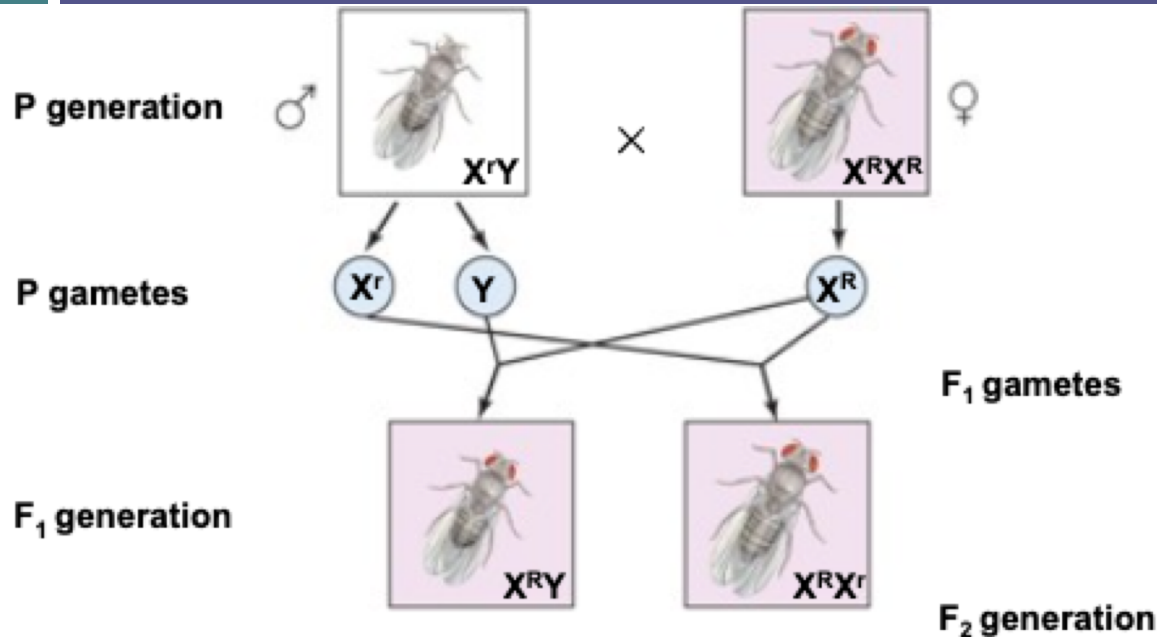
■ In mammals

- The X and Y chromosomes determine gender
- Females are XX; Males are XY

■ The term **X-linked** is used for genes that have nothing to do with gender

- X-linked genes are carried on the X chromosome.
- The Y chromosome does not carry these genes
- Discovered in the early 1900s by a group at Columbia University, headed by Thomas Hunt Morgan.
 - Performed experiments with fruit flies
 - They can be easily and inexpensively raised in simple laboratory glassware
 - Fruit flies have the same sex chromosome pattern as humans

X-Linked Inheritance



		eggs	
♀ ♂		X^R	X^r
sperm	X^R	 $X^R X^R$	 $X^R X^r$
	Y	 $X^R Y$	 $X^r Y$
		Offspring	

It was strange to Morgan's team that even though the inheritance followed the 3:1 of regular autosomal dominance, only males ever had white eyes!

Males that express a trait because they only receive one copy are called **Hemizygous**

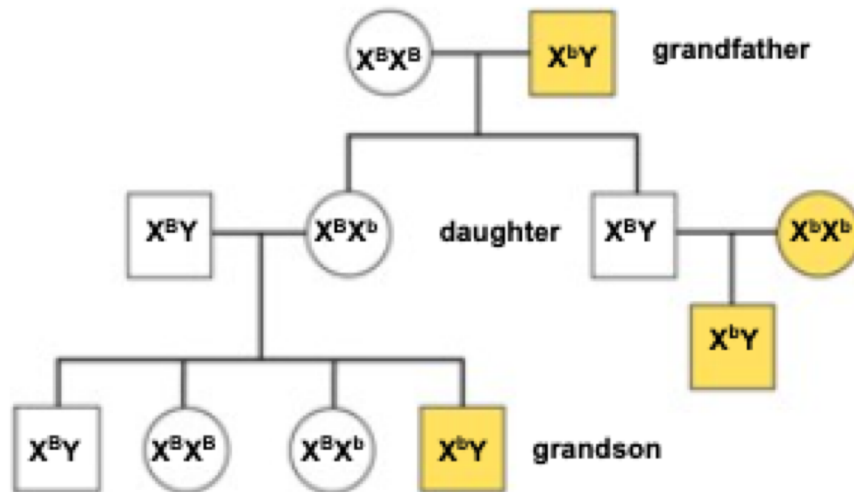
Allele Key
 X^R = red eyes
 X^r = white eyes

Phenotypic Ratio
 females: $\frac{3}{4}$ all red-eyed
 males: 1 $\frac{1}{2}$ red-eyed
 1 $\frac{1}{2}$ white-eyed

Extending the Range of Mendelian Genetics

- Several X-linked recessive disorders occur in humans:
 - ▣ Color blindness
 - The allele for the blue-sensitive protein is autosomal
 - The alleles for the red- and green-sensitive pigments are on the X chromosome.
 - ▣ Menkes syndrome
 - Caused by a defective allele on the X chromosome
 - Disrupts movement of the metal copper in and out of cells.
 - Phenotypes include kinky hair, poor muscle tone, seizures, and low body temperature
 - ▣ Muscular dystrophy
 - Wasting away of the muscle
 - Caused by the absence of the muscle protein dystrophin
 - ▣ Adrenoleukodystrophy
 - X-linked recessive disorder
 - Failure of a carrier protein to move either an enzyme or very long chain fatty acid into peroxisomes.
 - ▣ Hemophilia
 - Absence or minimal presence of clotting factor VIII or clotting factor IX
 - Affected person's blood either does not clot or clots very slowly.

X-linked Recessive Pedigree



Key	
$X^B X^B$	= Unaffected female
$X^B X^b$	= Carrier female
$X^b X^b$	= Color-blind female
$X^B Y$	= Unaffected male
$X^b Y$	= Color-blind male

X-Linked Recessive Disorders

- More males than females are affected.
- An affected son can have parents who have the normal phenotype.
- For a female to have the characteristic, her father must also have it. Her mother must have it or be a carrier.
- The characteristic often skips a generation from the grandfather to the grandson.
- If a woman has the characteristic, all of her sons will have it.

